Moon Mission Plant and Flight Software: Mathematics

1 Conventions

SI units throughout: meters, seconds, kilograms, radians. Frame: origin at Earth's center, z along Earth's rotation axis, x toward the vernal equinox, axes fixed relative to the stars (Earth-centered equatorial). Source: `workloads/moon/common/`.

2 Vectors

For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ and scalar s:

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z \tag{1}$$

$$\mathbf{a} \times \mathbf{b} = (a_y b_z - a_z b_y,\ a_z b_x - a_x b_z,\ a_x b_y - a_y b_x) \tag{2}$$

$$\|\mathbf{a}\| = \sqrt{\mathbf{a} \cdot \mathbf{a}} \tag{3}$$

The cross product follows the right-hand rule: $\hat{x} \times \hat{y} = \hat{z}$. Code: `vec3.hpp`. Tests: `test_vec3.cpp`.

3 Constants

Symbol	Value	Unit	Source
μ_E	$3.986004418 \times 10^{14}$	$\mathrm{m}^3/\mathrm{s}^2$	IERS Conventions 2010
μ_M	$4.902800066 \times 10^{12}$	$\mathrm{m}^3/\mathrm{s}^2$	JPL DE430
R_E	6 378 137	m	equatorial radius
R_M	1 737 400	m	mean radius
ω_E	7.292115×10^{-5}	rad/s	sidereal rotation
a_M	3.844×10^{8}	m	mean Earth–Moon distance
ε	23.439°		obliquity of the ecliptic
i_M	5.145°		lunar orbit inclination to the ecliptic
ϕ_L	28.573°		launch latitude, Kennedy LC-39A
g_0	9.80665	m/s^2	standard gravity

Code: `constants.hpp`.

4 Time

Mission time t is seconds since the epoch $\mathrm{JD}_0 = 2\,434\,743.5$ (1954-01-01 00:00 UTC; Bart Howard's original "Fly Me to the Moon" sheet music was published in 1954). Julian centuries from J2000:

$$T = \frac{(\mathrm{JD}_0 - 2\,451\,545.0) + t/86400}{36\,525} \tag{4}$$

The subtraction is evaluated first. $\mathrm{JD}_0 - 2\,451\,545.0 = -16\,801.5$ is exact in double precision, whereas adding $t/86400$ to JD_0 first rounds time to the spacing of doubles near 2.4×10^6, about 4×10^{-5} s, which made the differenced Moon velocity noisy at the 4×10^{-2} m/s level.

5 Moon ephemeris

Circular orbit of radius a_M. Mean elements, Meeus, *Astronomical Algorithms*, ch. 47, linear terms (higher-order terms change the result by about 0.0003° at the epoch):

$$L' = 218.3164477^\circ + 481\,267.88123421^\circ\, T \qquad \text{mean longitude} \tag{5}$$

$$\Omega = 125.0445479^\circ - 1934.1362891^\circ\, T \qquad \text{ascending node} \tag{6}$$

$$u = L' - \Omega \qquad \text{argument of latitude} \tag{7}$$

Position in the ecliptic frame:

$$\mathbf{r}_{\mathrm{ecl}} = a_M \begin{pmatrix} \cos\Omega\cos u - \sin\Omega\sin u\cos i_M \\ \sin\Omega\cos u + \cos\Omega\sin u\cos i_M \\ \sin u\,\sin i_M \end{pmatrix} \tag{8}$$

Rotation into the Earth-equatorial frame about x by ε:

$$\mathbf{r}_M = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\varepsilon & -\sin\varepsilon \\ 0 & \sin\varepsilon & \cos\varepsilon \end{pmatrix} \mathbf{r}_{\mathrm{ecl}} \tag{9}$$

The orbit's tilt I to Earth's equator follows from

$$\cos I = \cos\varepsilon\cos i_M - \sin\varepsilon\sin i_M\cos\Omega \tag{10}$$

At the epoch, $I = 25.988^\circ$, within the real range 18.3°–28.6° over the 18.6-year nodal cycle.

Limitations. The real orbit is eccentric ($e \approx 0.055$), so the modeled position differs from the true one by up to about 6° in longitude and 20 000 km in distance. $u = L' - \Omega$ is the standard approximation. Code: `moon_ephemeris.hpp`. Tests: `test_moon_ephemeris.cpp`.

5.1 Moon velocity

Central difference with $h = 1$ s:

$$\mathbf{v}_M(t) \approx \frac{\mathbf{r}_M(t+h) - \mathbf{r}_M(t-h)}{2h} \tag{11}$$

A smaller h amplifies rounding in $\mathbf{r}_M$; a larger h increases the truncation error from the path's curvature. Measured: speed 1023.18 m/s, matching the mean orbital speed $a_M\,\dot{L}'$; $\mathbf{r}_M \cdot \mathbf{v}_M$ vanishes to 3×10^{-11} of $\|\mathbf{r}_M\|\|\mathbf{v}_M\|$; the result differs from an $h = 0.5$ s difference by at most 5.2×10^{-4} m/s over 30 days, limited by rounding of the mean longitude angle (about 3900 rad at the epoch).

6 Navigation (flight software)

The flight software receives the Earth-centered state $(\mathbf{r}, \mathbf{v})$ and computes its own Moon state $\mathbf{r}_M(t)$, $\mathbf{v}_M(t)$. The Moon is tidally locked, so it spins at its orbital rate about its orbit normal:

$$\boldsymbol{\omega}_M = \frac{\mathbf{r}_M \times \mathbf{v}_M}{\|\mathbf{r}_M\|^2} \tag{12}$$

Its magnitude is $\dot{L}' + |\dot{\Omega}|(1 - \cos i_M)$: the regressing node adds a small rotation to the mean-longitude rate. Ground-relative state:

$$\mathbf{p} = \mathbf{r} - \mathbf{r}_M \qquad \text{Moon-centered position} \tag{13}$$

$$\mathbf{u} = (\mathbf{v} - \mathbf{v}_M) - \boldsymbol{\omega}_M \times \mathbf{p} \qquad \text{velocity relative to the ground} \tag{14}$$

$$\hat{\mathbf{n}} = \mathbf{p}/\|\mathbf{p}\| \qquad \text{local up} \tag{15}$$

$$h = \|\mathbf{p}\| - R_M \qquad \text{altitude} \tag{16}$$

$$v_\uparrow = \mathbf{u} \cdot \hat{\mathbf{n}} \qquad \text{vertical speed, positive up} \tag{17}$$

$$v_\leftrightarrow = \|\mathbf{u} - v_\uparrow \hat{\mathbf{n}}\| \qquad \text{horizontal speed} \tag{18}$$

This stage runs before guidance, so a fault here reaches every TMR replica.

Limitations. The real lunar spin axis is tilted about 1.5° from the orbit normal (Cassini's laws); the model aligns them. Altitude is above a sphere of radius R_M; there is no terrain. Code: `fsw/navigation.cpp`. Tests: `test_navigation.cpp`.

7 Guidance (flight software)

Inputs: the navigation state (Section on navigation) and the mass m. Output: throttle τ and direction $\mathbf{d}$; the plant applies the actuator limits.

Braking. $\tau = 1$, $\mathbf{d} = -\mathbf{u}/\|\mathbf{u}\|$: full thrust against the ground-relative velocity.

Switch. Braking ends, permanently, when $\|\mathbf{u}\| < 50$ m/s or $h < 2000$ m, whichever comes first.

Final descent. Target vertical speed and wanted acceleration:

$$v^* = -\operatorname{clamp}\left(\frac{h}{10\ \text{s}},\ 1,\ 30\right)\ \text{m/s} \tag{19}$$

$$g = \frac{\mu_M}{(R_M + h)^2} \tag{20}$$

$$\mathbf{a}^* = (g + k_v\,(v^* - v_\uparrow))\,\hat{\mathbf{n}} - k_h\,(\mathbf{u} - v_\uparrow \hat{\mathbf{n}}) \tag{21}$$

with $k_v = 1.0\ \text{s}^{-1}$ and $k_h = 0.5\ \text{s}^{-1}$. If $\mathbf{a}^* \cdot \hat{\mathbf{n}} \le 0$ the engine is off (a lander cannot thrust toward the ground). Otherwise

$$\tau = \frac{\|\mathbf{a}^*\|\,m}{F_{\max}}, \qquad \mathbf{d} = \mathbf{a}^* \tag{22}$$

and the plant saturates τ at 1. On the target descent rate with no drift, $\tau = g\,m/F_{\max}$, the hover throttle.

Initial values. The switch thresholds, the 10 s time constant, the 1 and 30 m/s limits, and both gains are starting values, to be tuned against the first full landing. Code: `fsw/guidance.cpp`. Tests: `test_guidance.cpp`.

8 Gravity

With $\mathbf{r}$ the spacecraft position, $\mathbf{r}_M$ the Moon position, and $r = \|\mathbf{r}\|$, $r_M = \|\mathbf{r}_M\|$:

$$\mathbf{g} = -\mu_E \frac{\mathbf{r}}{r^3} + \mu_M \left(\frac{\mathbf{r}_M - \mathbf{r}}{\|\mathbf{r}_M - \mathbf{r}\|^3} - \frac{\mathbf{r}_M}{r_M^3} \right) \tag{23}$$

The first term is Earth's pull. The Moon contribution is the third-body perturbation: the Moon's pull on the spacecraft minus the Moon's pull on Earth. The subtracted part, the indirect term, is required because the frame is centered on Earth, which itself accelerates toward the Moon. Near Earth the two Moon terms nearly cancel; near the Moon the first dominates. Code: `dynamics.cpp`. Tests: `test_dynamics.cpp`.

9 Engine and actuator limits

The flight software commands a throttle τ and a direction $\mathbf{d}$. The plant applies the limits of a real engine:

$$\tau' = \begin{cases} 1 & \tau > 1 \\ \tau & 0 < \tau \le 1 \\ 0 & \text{otherwise (including NaN)} \end{cases} \tag{24}$$

The engine is off ($\tau' = 0$) when $\|\mathbf{d}\|$ is zero, infinite, or NaN, or when the mass has reached the dry mass m_{dry}. Otherwise:

$$F = \tau' F_{\max} \tag{25}$$

$$\mathbf{a}_T = \frac{F}{m} \frac{\mathbf{d}}{\|\mathbf{d}\|} \tag{26}$$

$$\dot{m} = -\frac{F}{I_{sp}\, g_0} \tag{27}$$

The direction is normalized, so a corrupted command can point the engine anywhere but cannot push harder than full thrust. These limits are physics of the plant, not a flight software defense.

10 Equations of motion

$$\dot{\mathbf{r}} = \mathbf{v} \tag{28}$$

$$\dot{\mathbf{v}} = \mathbf{g}(\mathbf{r}, t) + \mathbf{a}_T \tag{29}$$

$$\dot{m} = -\frac{F}{I_{sp}\, g_0} \tag{30}$$

Code: `dynamics.cpp`, function `derivative`.

11 Numerical integration (RK4)

Write the state as $\mathbf{y} = (\mathbf{r}, \mathbf{v}, m)$ and the equations of motion as $\dot{\mathbf{y}} = f(\mathbf{y}, t)$. The classical fourth-order Runge–Kutta step of size h:

$$\mathbf{k}_1 = f(\mathbf{y}_n,\, t_n) \tag{31}$$

$$\mathbf{k}_2 = f(\mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_1,\, t_n + \tfrac{h}{2}) \tag{32}$$

$$\mathbf{k}_3 = f(\mathbf{y}_n + \tfrac{h}{2}\mathbf{k}_2,\, t_n + \tfrac{h}{2}) \tag{33}$$

$$\mathbf{k}_4 = f(\mathbf{y}_n + h\,\mathbf{k}_3,\, t_n + h) \tag{34}$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \tfrac{h}{6}\,(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4) \tag{35}$$

The engine command is held constant for the whole step, as flight software sets one command per tick. If $m_{n+1} < m_{\text{dry}}$, it is set to m_{dry}; the step still applied thrust for its full length, so the error is at most one step of thrust.

Validation. With no thrust, specific orbital energy $\mathcal{E} = \frac{1}{2}\|\mathbf{v}\|^2 - \mu_E/r$ and angular momentum $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ are conserved. Over one 400 km circular orbit in 5000 steps, measured drift is 2.5×10^{-9} in $\mathcal{E}$ and 1.3×10^{-9} in $\|\mathbf{h}\|$, with the Moon's tidal pull present; the spacecraft returns within 3 m of its start. Forward Euler on the same orbit drifts 1.5×10^{-2} in $\mathcal{E}$ and returns 510 km off. Code: `integrator.cpp`. Tests: `test_integrator.cpp`.